

Job Sequencing

and Tool Switching

A grouping gap, then an exact Benders study

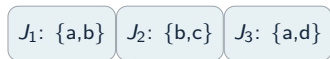
Shankar

Internship defence



Why does the job order matter?

The machine can hold two tools. It loads missing tools before a job; a load costs one.



$J_1 \rightarrow J_2 \rightarrow J_3$: after free fill, load c , then a, d
 $\Rightarrow$ 3 paid insertions

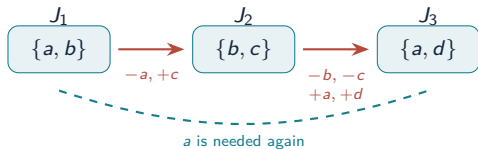


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Different order: $J_2 \rightarrow J_1 \rightarrow J_3$ needs only **2** insertions.

Model: job requirements T_j , capacity b ; initial loading is free.

For one fixed order, KTNS gives the best loading



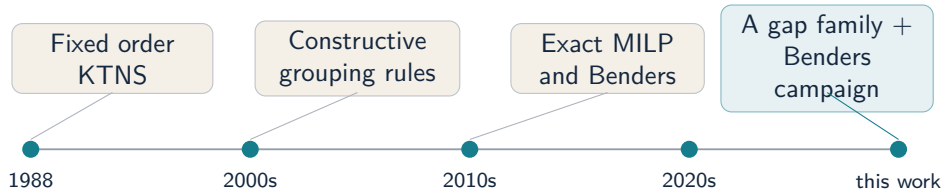
KTNS rule

When space is needed, remove the currently loaded tool whose next use is furthest in the future.

For a fixed order, this is optimal and polynomial-time. The hard decision is the job order.

Tang and Denardo (1988); standard KTNS interpretation.

Where this work sits



Implemented baselines

LSS, CATZ-F4, SSPMF.

Scope

Reimplementations for a controlled comparison, not a claim of a complete SOTA benchmark.

Literature review and references: report, Chapters 2–3.

The grouping heuristic commits before ordering



K^*

Fewest feasible groups.
This is what the grouping phase minimises.

Z^*

Best KTNS cost over every job order.

H

Best KTNS cost among orders induced by a K^* -grouping.

Definitions used throughout the theoretical part of the report.

Every cost is a baseline plus reinsertions

$$\text{cost} = q + R$$

q

first insertion of each
tool

$q = |U| - b$ with a free
full magazine.

R

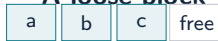
returns of tools
loaded before

The grouping gap is
entirely in R .

A tight block



A loose block



A spare slot lets a useful tool stay resident
across the block.

Coverage accounting lemma; report Chapter 4.

A four-job counterexample: grouping already loses one

	A	B	C	D
a	●		●	
b	●			●
c		●	●	
d		●		
e			●	●
f			●	●
g				●

Capacity $b = 4$. Only A and B fit in one magazine together, so grouping must pair them.

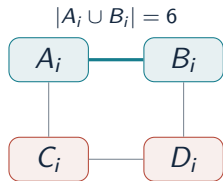
Unrestricted: B, C, A, D costs $3 = q$.

Grouped: every group-respecting order costs at least 4.

The loss is not a tie-breaking artefact: all six group orders can be checked.

Seed instance used to motivate the infinite construction; report Chapter 4.

A connected family forces the same grouping decision



u occurs in every job, so all copies are connected. The private marker v_i identifies copy i .

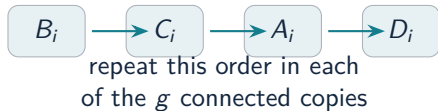
Set $b = 6$. Within a copy, A_i, B_i are the only pair that fits together. Two small jobs from different copies use 7 tools.

Therefore the grouping is unique, and

$$K^* = 3g.$$

Family $\mathcal{F}_g$: construction and verification table in report Chapter 4.

Unrestricted ordering reaches the lower bound



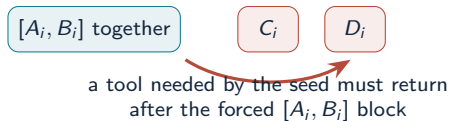
The first copy needs 3 paid insertions.
Each later copy needs 8.

$$Z^* = 8g - 5 = q$$

The schedule exactly meets the
coverage lower bound.

Upper schedule plus coverage bound; report Chapter 4.

The forced grouping pays one extra return per copy



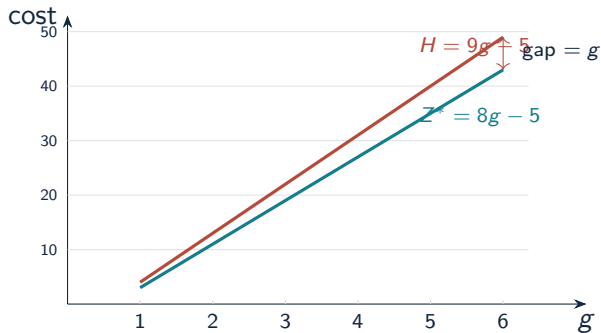
Because u and v_i stay present, the four remaining slots are fully occupied at the critical boundary.

Each copy contributes one forced reinsertion:

$$H = q + g = 9g - 5.$$

Forced-return lemma and matching schedule; report Chapter 4.

The additive loss grows linearly



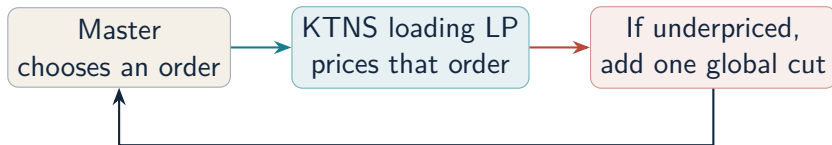
Result

The restricted family has an unbounded additive gap.

The ratio tends to $9/8$; this construction does not prove a universal ratio bound.

Theorem: $H - Z^* = g$ for $g \geq 1$; report Chapter 4.

One Benders iteration: order first, exact cost second

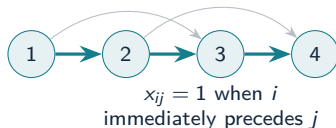


The incumbent is always a feasible order checked by KTNS.

Separation of roles: combinatorial ordering in the master; exact fixed-order loading in the oracle.

Logic-based Benders architecture used in the BBC configurations.

The master chooses a Hamiltonian path



$$\min \theta, \quad \sum_{i \neq j} x_{ij} = 1, \quad \sum_{j \neq i} x_{ij} = 1$$

$$\sum_{i,j \in S} x_{ij} \leq |S| - 1 \quad (S \subseteq J)$$

$$\theta \geq |U|, \quad \theta \geq \sum_{i,j \in J} w_{ij} x_{ij} + \sum_{j \in J} |T_j| x_{dj}.$$

A dummy depot d marks the start and end. The model uses empty-start cost; subtract b to recover the free-fill convention. Fractional SEC separation was not tested.

Master formulation: report Chapter 5.

The loading oracle prices one path exactly

$$\begin{aligned} \min \quad & \sum_{j,t} z_{jt} \\ \text{s.t.} \quad & y_{jt} \geq 1 && (t \in T_j), \\ & \sum_t y_{jt} \leq b && (j \in J), \\ & z_{jt} \geq y_{jt} - y_{it} - (1 - x_{ij}) && (i \neq j, t \in U), \\ & z_{jt} \geq y_{jt} - (1 - x_{dj}) && (j \in J, t \in U), \\ & y, z \geq 0. \end{aligned}$$

y_{jt} says tool t is loaded while job j runs; z_{jt} charges its insertion.

For a fixed path, the LP is integral and its optimum equals KTNS.

Polynomial oracle

Fixed-order loading formulation and integrality argument: report Chapter 5.

One dual solution gives a global cut

$$\theta \geq \sum_{i \neq j} \sum_t \bar{\lambda}_{ijt} (x_{ij} - 1) + \sum_j \sum_t \bar{\lambda}_{jt}^d (x_{dj} - 1) - b \sum_j \bar{\mu}_j + \sum_j \sum_{t \in T_j} \bar{\nu}_{jt}.$$

Why it is global

The dual feasible region does not depend on the proposed order. The row is valid for every x and tight at the order just priced.

What it learns

Immediate adjacency prices.

What it misses

Some longer residency effects only emerge after branching.

Dual-derived Benders cut; derivation and scope: report Chapter 5.

What was compared, under one common budget

LP	dual optimality cuts at integer orders
+T	triplet lower bounds
+F	Benders cuts at fractional nodes
+H	hybrid-genetic incumbent and seed cut
+C	conflict-graph lower bound
+P	Pareto-optimal cut lifting

Protocol

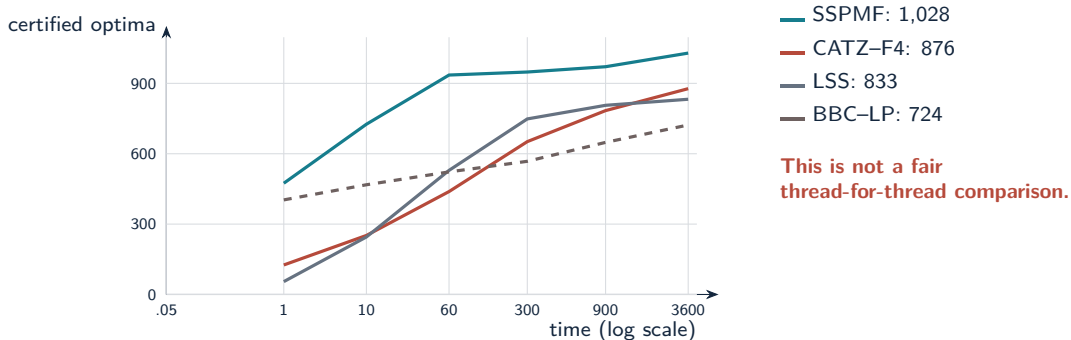
1,410 canonical instances; 12 configurations; one-hour cap; 16 GiB.

BBC used 1 CPLEX thread; compact baselines used 4.

+F disables presolve. +H includes a seed cut and about two seconds of uncharged preparation.

Complete 16,920-row campaign; reproducibility ledger in report Appendix C.

Cactus view: solve counts over time



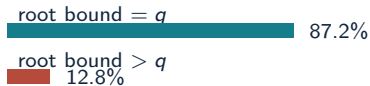
Full campaign: 1,410 instances, 3,600 seconds; BBC 1 thread, compact baselines 4.

Coverage gap separates the methods

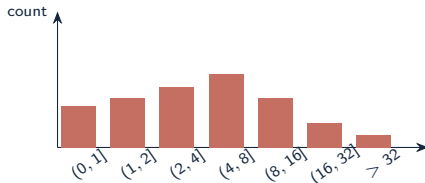


Solved share by discrete coverage-gap bin; 303 cases excluded because the optimum is unknown.

Most BBC root bounds stop at the coverage bound



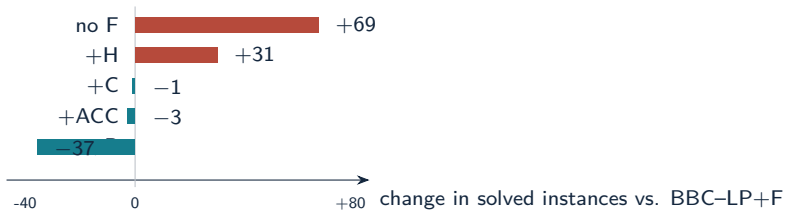
9,067 / 10,392 root records equal q ; records are pooled over nine BBC regimes.



Among unfinished known-opt BBC-LP runs, 223/383 incumbents are optimal; median final gap is 23.1%.

Root-bound audit and incumbent audit; report Chapter 6.

Ablation: more cuts did not mean more solves



BBC-LP+F solves 655.

Pareto lifting gives 618 (-37).

Fractional cuts reduce median nodes 77,439 → 3,701, yet lose 69 solves.

Interpret with care: +F also disables presolve; +H includes a seed cut.

BBC component ablation, 1-hour cap; report Chapter 6.

What this project establishes, and what it does not

Theory

A connected family where grouping incurs g extra insertions.

Computation

A reproducible, complete campaign that exposes a weak root-bound regime.

Next test

Cuts that price longer residency spans, then a fairer parallel comparison.

The evidence supports a focused next algorithmic question, not a SOTA claim.

Synthesis of Chapters 4–7; limitations stated explicitly in the report.

Backup: proof ledger for the grouping-gap theorem

Claim	Certificate in the report
$K^* = 3g$	Unique feasible pairs identify the forced grouping.
$Z^* = 8g - 5$	Coverage lower bound q plus an explicit unrestricted schedule.
$H = 9g - 5$	A forced return at each copy plus a matching grouped schedule.
Connectedness	Universal tool u links every copy.

Scope: this proves the stated family and heuristic restriction; it does not settle a universal grouping approximation ratio.

Full line-by-line argument: report Chapter 4 and THEOREM_AUDIT.md.

Backup: campaign provenance and comparison discipline

Item	Recorded decision
Instances	1,410 canonical instances; 12 configurations; 16,920 completed rows.
Budget	3,600 seconds and 16 GiB per configuration.
Parallelism	BBC: 1 CPLEX thread. Compact baselines: 4 threads.
Fairness	Baselines were reimplemented; compare evidence, not a universal SOTA ranking.
Artifacts	Canonical results and minimal OOM evidence are public; the 289 MB duplicate server tree is ignored.

Reproducibility ledger: report Appendix C; public-repository hygiene note.

Backup: configuration models rarely lift the coverage bound

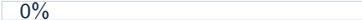
PCF': configurations by position

Its LP relaxation equals q by proof. In the campaign, **0 of 2,108** recorded roots exceeded q .

PTF: transitions between configurations

Its LP can be stronger. It exceeded q on only **2 of 232** recorded roots.

share of roots above q

PCF'  0%

PTF  0.9%

On the two exceptions, the root rises from 5 to 6 while the optima are 12 and 13.

Branch-and-price root-bound audit; exact values and definitions in report Chapter 6.